HelpingHand

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Introduction

Purpose / Need

The purpose of Helping Hand is to further bridge a gap between American Sign Language (ASL) students and the Deaf community. While many resources exist for visual learning, beginners often lack immediate, tactile feedback for handshapes and movements. This project fulfills a need for hearing learners by providing a wearable glove with real-time haptic feedback to correct finger positioning during practice. Beyond a mere demonstration, Helping Hand provides structured learning through a companion mobile app that provides modules for basic letters and numbers.

Domain & Prior Art

The project involves wearable technology, haptic interfaces, and machine learning. While prior work like SignAloud from UW utilized flex sensors and IMUs for speech-to-text translation via Bluetooth or the UCLA smart glove with a library of 660 signs, our design acts as an active educational tool that provides corrective haptic feedback to the wearer. While recent studies on Chinese Sign Language dataglovers demonstrate the efficiency of inertial sensors for social access, our project integrates a mobile app dashboard to track long-term student progress. Furthermore, where earlier systems rely on PIC microcontrollers and lookup tables for computer displays, Helping Hand uses the ESP32 for neural network classification and wireless connectivity. Finally, unlike two-way communication systems that require external cameras and audio input for recognition, our solution is self-contained in the glove and provides portable, physical vibration patterns for error.

Impact & Risk Assessment

HelpingHand fosters social integration by providing hearing beginners with haptic feedback to build muscle memory for ASL, serving as a linguistic bridge that respects the Deaf community's preferred communication modes. Economically, the device offers a low-cost educational supplement, while its modular design minimizes electronic waste by allowing for specific component repairs. However, ethical risks include potential social confusion from

machine learning misinterpretations and the technical inability to capture non-manual markers like facial expressions, which are fundamental to ASL grammar. While the system is currently restricted to beginner vocabulary and susceptible to sensor fatigue, it remains an effective corrective tool designed to facilitate early-stage learning rather than replace the cultural and linguistic depth of human-to-human conversation.

Statement of Work

HelpingHand uses a dual environment software architecture to manage real-time data processing and user interaction. The embedded software on the ESP32 will be developed in C using the React Native framework, utilizing the NimBLE host stack to manage Bluetooth Low Energy (BLE) communication and optimize power consumption while maintaining a stable Generic Attribute Profile (GATT) connection for streaming sensor data real-time with a throughput requirement of approximately 2-5kbps. The dependencies may include react-native-ble-manager for device communication and Victory Native or similar libraries for progress visualization. The hardware layer consists of five resistive flex sensors, sampled using at least 10-bit ADC, to detect handshape and a wrist-mounted 6-axis IMU, interfaced using I2C at 400kHz, to track movement path, speed, and orientation. This configuration allows the system to capture the primary parameters of ASL: handshape, movement, and orientation. The mobile application, developed in React Native, serves as the central hub for user education and gesture recognition. It will utilize a neural network classifier trained on ASL datasets to map incoming sensor data to specific characters with a target accuracy of about 90%.

During the first semester, the software scope focuses on the ASL alphabet (A-Z) and numbers (0-9), while the second semester introduces more complex dynamic signs, such as “HELLO” and “THANK YOU.” This dual-semester roadmap includes integrating a local SQLite database to track performance trends and conducting user outreach with the UF ASL community to validate the effectiveness of the haptic learning model.

Deliverable Artifacts

Hardware: The HelpingHand Smart Glove

The primary physical deliverable is a wearable smart glove prototype integrated with an ESP32 microcontroller, five flex sensors, and a 6-axis IMU. This artifact serves as the data acquisition hub, capturing the five key parameters of ASL, handshape, movement, and orientation while providing real-time haptic feedback via vibration motors to the user’s fingertips. To ensure long-term usability and maintenance, the glove features a modular design where sensors are housed in replaceable fabric channels and the electronics are protected by a wrist enclosure. This allows for the physical components to be repaired or upgraded without discarding the entire system.

Software: ASL Gesture Recognition Engine & Training Data

HelpingHand will deliver a trained Machine Learning (ML) model, likely a neural network or LSTM, capable of classifying the ASL alphabet, numbers 0-9, and a set of core beginner signs. This model acts as the interpretive layer of the project, processing raw sensor data into recognized text for the companion application. The software artifact includes the source code for training, the final model, weights, and the custom datasets recorded during development. All software will be disseminated via a GitHub repository, including a README with documentation for the communication protocol, allowing others to build upon our recognition logic.

Software: HelpingHand Companion Application

A mobile application will provide a user-facing interface that handles BLE communication with the glove. This app displays recognized signs and provides a calibration interface for the user. The dissemination plan involves hosting the source code on GitHub alongside a pre-compiled executable (APK) for immediate testing. To maintain viability, the app is designed with an extensible UI that can support the add-on of new signs as the ML model expands, ensuring that the software remains useful beyond the scope of this class.

Documentation: Technical Specification and User Manual

The final deliverable is documentation consisting of a user manual, circuit schematics, and a maintenance report. This documentation provides the “blueprints” for the project, detailing the flex sensors and the I2C communication protocols used by the IMU. These documents will be formatted as PDF reports within the version control system. By documenting the BLE API calls and calibration procedures, it ensures that our system is maintainable and usable for future users or engineers.

Mockups

Storyboard

Panel 1: The user pairing the glove and application using Bluetooth with the microcontroller broadcasting the device UUID. The app establishes connection with sensor characteristics.

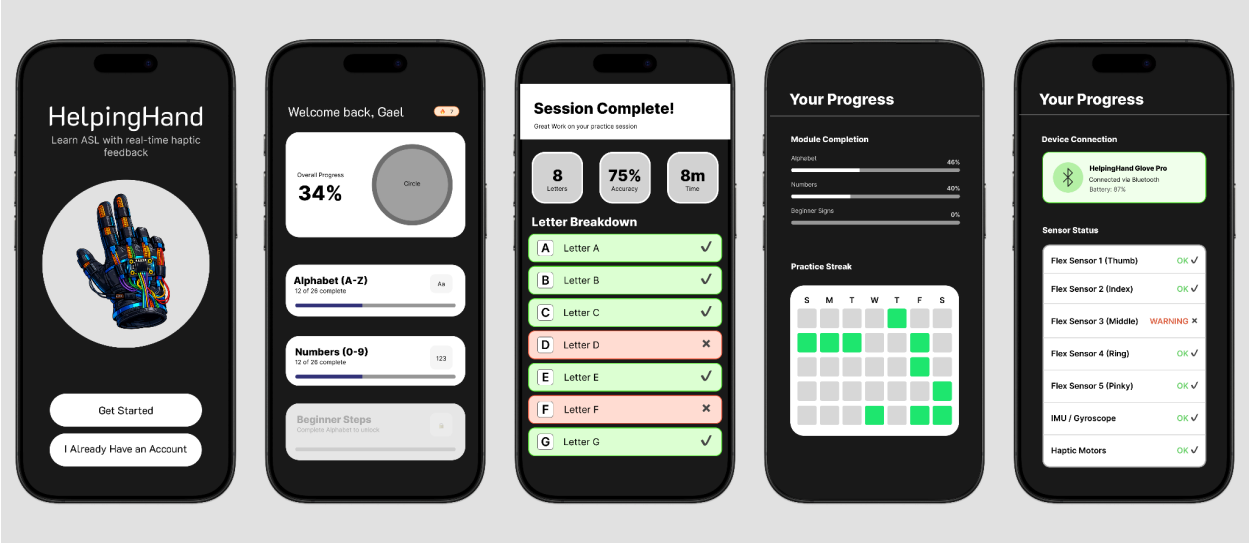
Panel 2: The user selects the alphabet module and chooses to practice the letter ‘B’, which is signalled to the glove. In turn, the glove streams real-time flex sensor ADC values at a proper sampling rate to app for gesture recognition.

Panel 3: The on-glove gesture classifier compares the live sensor array against the stored B letter template with a close-up shot of the flex sensors. For reference, the letter ‘B’ has the thumb folded over the palm instead of sticking out as seen in the panel.

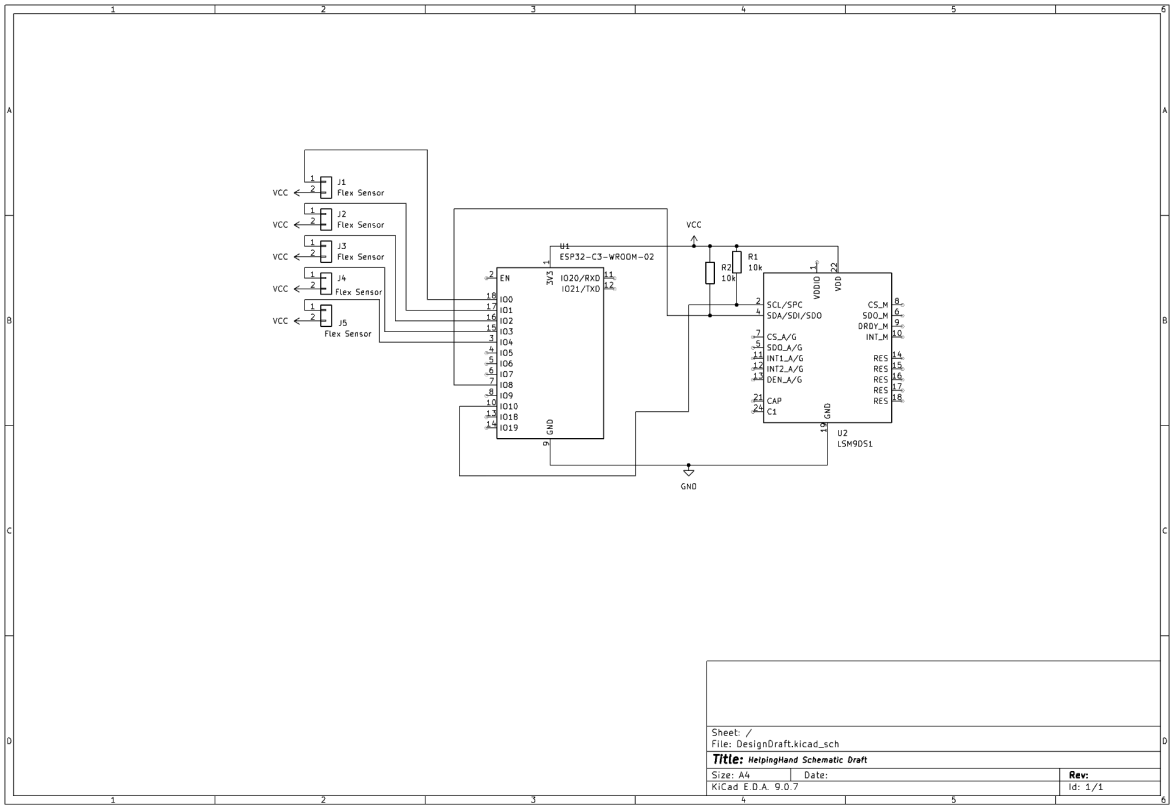
Panel 4: The glove receives correction commands via Bluetooth and activates its vibration motors. Meanwhile, the app renders real-time sensor data visualization with error highlighting.

Panel 5: The app aggregates session metrics from an online database. The program calculates accuracy percentage, generates performance analytics, and syncs anonymized learning data to cloud backend.

Figma Design of App



Schematic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J2
Flex Sensor
J3
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J4
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J5
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VCC
U1
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EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
3V3
IO20/RXD
IO21/TXD
GND
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10k
R2
10k
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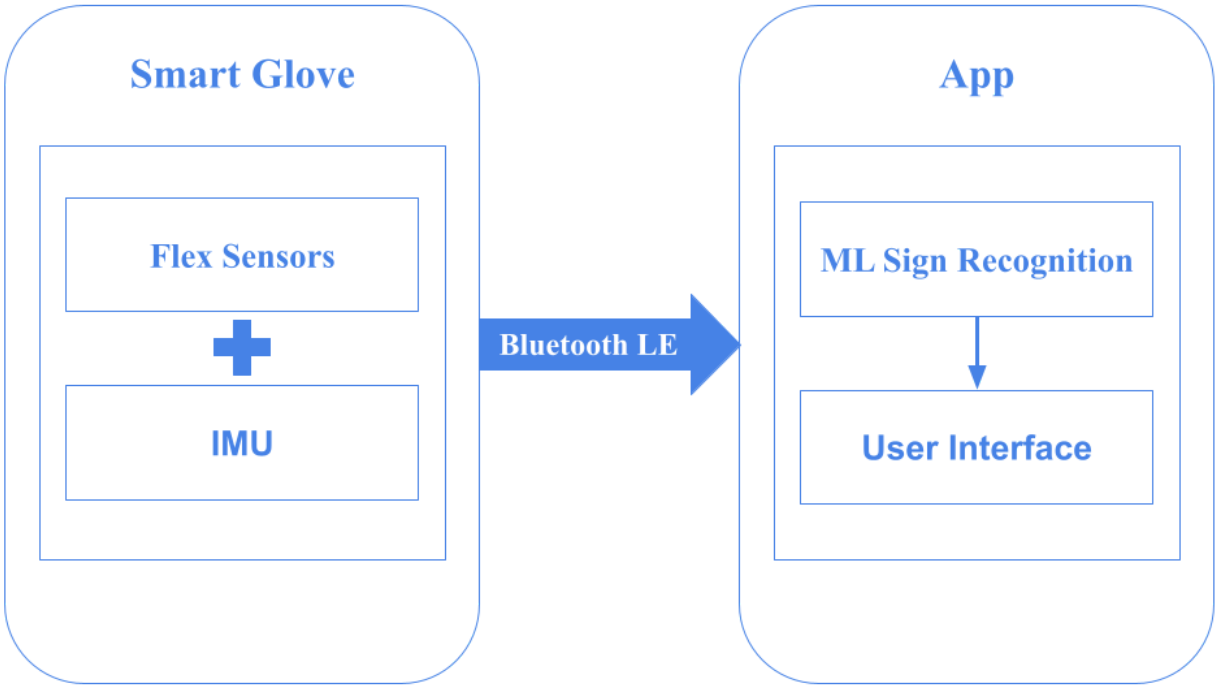
Networking Diagram



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